

M. R. Tanush Pai

AI/ML Engineer · Generative AI & LLM Systems · Agentic AI · Python Backend Engineering
tanushpai06@gmail.com | +91-9567805222 | Kerala, India | linkedin.com/in/tanushpai | github.com/tanushpai

PROFESSIONAL SUMMARY

AI/ML Engineer specializing in production-grade Generative AI systems, agentic workflows, and intelligent data platforms. Experienced building LLM-powered analytics assistants, enterprise document generation systems, and AI solutions that automate complex business workflows. Strong background in Python backend development, FastAPI, PostgreSQL, and scalable AI application architecture.

TECHNICAL SKILLS

Languages: Python, SQL, R
AI/ML & GenAI: Scikit-learn, LangChain, LangGraph, Hugging Face, OpenAI API, Retrieval-Augmented Generation (RAG), Agentic Workflows, Ollama
Data Processing: Pandas, NumPy, PySpark
Databases, ORM & Vector Stores: PostgreSQL, MongoDB, Redis, Prisma, FAISS, Vector Search
Deployment & Version Control: Vercel, Git, GitHub, Bitbucket
Project Management: Jira

PROFESSIONAL EXPERIENCE

AI/ML Engineer Intern | Sam Corporate Kochi, India
Feb 2026 – Present

- Designed and implemented the end-to-end architecture for an agentic AI analytics chatbot with an LLM-based planner, dynamically generating MongoDB aggregation pipelines across 25+ collections while ensuring GDPR-aligned data privacy and secure query execution.
- Built an AI-powered QMS document generation platform adopted by 12+ employees, leveraging LLMs to generate content from user prompts across 10+ templates, reducing document creation time from hours to minutes through automated formatting and content validation.
- Enhanced an AI-powered ESG analytics platform by implementing new features across 8+ sustainability analysis modules, including AI-driven report generation, benchmarking, peer analysis, NL2SQL, and interactive chart generation using PostgreSQL, OpenAI, and Perplexity APIs.
- Resolved a production-critical LLM reliability issue by optimizing prompt size and API request workflows, eliminating HTTP 429 rate-limiting errors and improving application stability.
- Built an automated MongoDB data validation tool that compared records across multiple databases and generated Excel reports, replacing manual spreadsheet-based verification with a single-click workflow.

Data Analytics Intern | Navodita Infotech Kerala, India
Aug 2025 – Sep 2025

- Developed a user-based collaborative filtering recommendation system using Python, Pandas, and NumPy by preprocessing 10K+ user-movie ratings, implementing cosine similarity, and generating Top-10 personalized movie recommendations.

Data Science Intern | Keltron REC Kerala, India
Jun 2025

- Performed data preprocessing on 5+ datasets, created Power BI dashboards, and gained hands-on experience in Excel, SQL, and Machine Learning fundamentals.

KEY PROJECTS

AI Chatbot – Agentic Analytics & RAG Assistant | *Python, FastAPI, OpenAI API, FAISS, MongoDB, Redis*

- Built an agentic analytics assistant that lets staff type plain-English questions and receive validated charts or insights, eliminating the need to write MongoDB queries manually.
- Designed the end-to-end architecture and core modules for the analytics assistant, defining the workflow for query planning, validation, and response generation.
- Implemented an LLM-based AI planner to replace traditional intent classification, enabling robust handling of complex user queries through dynamic reasoning and task planning.
- Engineered dynamic AI-generated MongoDB aggregation pipeline generation across 25+ collections, enabling non-technical users to retrieve validated analytics and chart-ready insights using natural language.

AVANA – Autonomous Visual Analytics & Normative Agents | *Python, LangChain, LangGraph, Ollama*

- Architected a 3-agent LangGraph pipeline (*Data Engineer* → *Data Analyst* → *Data Scientist*) that transforms raw CSV uploads into cleaned datasets, exploratory data analysis reports, and trained machine learning models with zero manual intervention.
- Implemented a Master Orchestrator using a shared Dataset Contract (target column, problem type, domain context) to coordinate agents across a 13-stage data preprocessing pipeline, 50+ automated visualizations, and FLAML AutoML training.
- Built an AI Consultant chat interface powered by Ollama (Llama 3.1) with role-aware personas, enabling contextual interaction with pipeline outputs, visualizations, and model insights.

AcademicXchange | *HTML, CSS, JavaScript, Python, Hugging Face*

- Developed a responsive educational platform providing centralized access to notes, syllabus, and previous question papers for KTU students.
- Implemented an AI-powered Question Paper Generator that automatically creates question papers based on selected topics and syllabus coverage.
- Built an AI-powered PDF Summarizer and interactive Study Guide chatbot using Hugging Face models to generate concise summaries, answer queries, and create practice questions.

EDUCATION

SCMS School of Engineering and Technology

B.Tech in Computer Science (Data Science)

Kerala, India

Oct 2022 – Apr 2026

CERTIFICATIONS

- Data Analysis with Python and Pandas – Infosys Springboard (2025)
- Python for Data Science – Infosys Springboard (2025)
- Multi AI Agent Systems with CrewAI – DeepLearning.AI, Coursera (2025)
- Fundamentals of Building AI Agents – IBM, Coursera (2025)